

AD 502 – Machine Learning

Pre-Reading Material – Lecture 8

Hypothesis Testing

Estimated Reading Time: 25–30 minutes

Why This Topic Matters

Machine learning and data analysis often require us to determine whether an observed difference or relationship is meaningful or simply due to random variation. **Hypothesis testing** provides a statistical framework for making such decisions using sample data and helps support evidence-based conclusions.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the purpose of **hypothesis testing**.
- Differentiate between the **null hypothesis** and **alternative hypothesis**.
- Understand **significance level, p-value, and test statistic**.
- Explain **Type I and Type II errors**.
- Distinguish between **one-tailed and two-tailed tests**.
- Understand the basic idea of common statistical tests.

Activate Your Prior Knowledge

1. Sample vs Population

Suppose you want to know the average marks of all students in a college but cannot ask every student.

Think: How can a sample be used to make a conclusion about the entire population?



2. Is the Difference Real?

Suppose Model A achieves 92% accuracy while Model B achieves 89%.

Think: Is Model A actually better, or could this difference have occurred because of random variation?

3. Making a Decision

Imagine a company claims that its new algorithm improves prediction accuracy.

Think: What statistical evidence would you need before accepting the claim?

Prerequisite Concepts

You should be familiar with:

- Population and sample
- Mean and variance
- Probability
- Normal distribution
- Sampling
- Basic descriptive statistics

Core Concepts

1. Hypothesis Testing

Hypothesis testing is a statistical method used to evaluate a claim about a population using sample data. It helps determine whether there is sufficient evidence to reject a proposed assumption.



2. Statistical Hypothesis

A statistical hypothesis is a statement about a population parameter or distribution. Hypothesis testing evaluates whether the available sample provides enough evidence to support or reject that statement.

3. Null Hypothesis (H_0)

The **null hypothesis** represents the default assumption, usually stating that there is no significant difference, effect, or relationship. It is the hypothesis tested directly using statistical evidence.

4. Alternative Hypothesis (H_1 or H_a)

The **alternative hypothesis** represents the claim that a significant difference, effect, or relationship exists. It is considered when sufficient evidence is found against the null hypothesis.

5. Example of Hypotheses

Suppose we want to determine whether a new ML algorithm improves accuracy.

- **H_0 :** The new algorithm does not improve accuracy.
- **H_a :** The new algorithm improves accuracy.

The sample results are then used to decide whether H_0 should be rejected.

6. Significance Level (α)

The significance level is the maximum probability of rejecting a true null hypothesis that we are willing to accept. Common choices include **0.05** and **0.01**.

7. P-Value

The **p-value** indicates how compatible the observed sample result is with the null hypothesis. A small p-value provides stronger evidence against H_0 .

8. Decision Rule

A common decision rule is:

- If **p-value** $\leq \alpha$, reject H_0 .
- If **p-value** $> \alpha$, fail to reject H_0 .

Failing to reject H_0 does **not** prove that the null hypothesis is true.

9. Test Statistic

A test statistic is a numerical value calculated from sample data. It measures how far the observed result is from what would be expected under the null hypothesis.

10. One-Tailed Test

A one-tailed test examines whether a parameter is significantly **greater than or less than** a specified value. The direction of the alternative hypothesis determines the tail being tested.

11. Two-Tailed Test

A two-tailed test examines whether a parameter is significantly **different from** a specified value in either direction. Evidence in both tails of the probability distribution is considered.

12. Type I Error

A **Type I error** occurs when the null hypothesis is rejected even though it is actually true. Its probability is represented by the significance level α .

13. Type II Error

A **Type II error** occurs when the null hypothesis is not rejected even though the alternative hypothesis is actually true. Its probability is commonly represented by β .

14. Statistical Power

Statistical power is the probability of correctly rejecting a false null hypothesis. It is given by $1-\beta$ and increases when a test has greater ability to detect a real effect.

15. Z-Test and T-Test

A **z-test** is commonly used when the population standard deviation is known or when appropriate large-sample assumptions apply. A **t-test** is commonly used when the population standard deviation is unknown, particularly with smaller samples.

16. Chi-Square Test

The **chi-square test** is commonly used for categorical data, such as testing relationships between categorical variables or comparing observed and expected frequencies.

17. ANOVA

Analysis of Variance (ANOVA) is used to determine whether there are statistically significant differences among the means of multiple groups. It extends the basic idea of comparing means beyond two groups.



18. Hypothesis Testing in Machine Learning

Hypothesis testing can be used to compare ML models, evaluate whether an observed improvement is statistically significant, and determine whether relationships found in data are likely to reflect more than random variation.

Guided Reading Questions

1. What is the purpose of hypothesis testing?
2. What is the difference between H_0 and H_a ?
3. What does a p-value tell us?
4. What is the role of the significance level α ?
5. What is the difference between rejecting and failing to reject H_0 ?
6. What is the difference between Type I and Type II errors?
7. When would a one-tailed test be appropriate?
8. What is the difference between a z-test and a t-test?
9. When is a chi-square test useful?
10. How can hypothesis testing be used to compare ML models?

Reflection Activity

Write 100–150 words explaining how hypothesis testing could be used to determine whether a new machine learning model is **significantly better** than an existing model. Consider the role of the null hypothesis, alternative hypothesis, p-value, and significance level.

Self-Assessment

1. The default assumption in hypothesis testing is called the _____ hypothesis.
2. The hypothesis representing the presence of an effect is called the _____ hypothesis.

3. A commonly used significance level is:

A. 0.5

B. 0.05

C. 5

D. 50

4. If $p\text{-value} \leq \alpha$, we generally:

A. Accept H_0 as true

B. Reject H_0

C. Increase the sample mean

D. Reject the data

5. Rejecting a true null hypothesis is called a _____ error.

6. Failing to reject a false null hypothesis is called a _____ error.

7. Statistical power is represented by:

A. α

B. β

C. $1-\beta$

D. $1-\alpha$

8. A test that considers deviations in both directions is called a _____ test.
9. A chi-square test is commonly associated with _____ data.
10. ANOVA is commonly used to compare the means of _____ groups.

Key Takeaway

Hypothesis testing provides a systematic way to determine whether evidence from sample data is strong enough to support a claim. Understanding **H₀**, **H_a**, **p-value**, **significance level**, **errors**, and **statistical tests** is essential for making reliable conclusions from data and evaluating machine learning results.

